

Association-Rules Market-Basket Recommender

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Assignment: HW4 — Association Rules (Week 4)

Abstract

We implement an Apriori-based association-rule miner that runs entirely in the browser over the UCI *Online Retail* dataset, cleaned to 17,080 baskets and 3,653 distinct items, and use it to answer the assignment's business and algorithmic analysis questions. At the final operating point (support $\geq 1\%$, confidence $\geq 30\%$) the miner produces 1,219 frequent itemsets and 950 rules, all with lift > 1 . A cross-sell rule between two pieces of the same china pattern (GREEN $\rightarrow$ ROSES REGENCY TEACUP, confidence 78.29%, lift 17.1440) is analysed as a deployable recommendation, while a rule into the most frequent item in the catalogue (HOME BUILDING BLOCK WORD $\rightarrow$ WHITE HANGING HEART T-LIGHT HOLDER, confidence 30.29%, lift 2.6409) illustrates how confidence alone misleads: at a 30% confidence floor no rule into the top item can score below $0.30 / 0.1147 = 2.62$ lift, so passing the filter is close to automatic. Threshold sweeps from 0.5%/30% to 3%/60% produce between 10,723 and 5 rules and are explained through downward closure. We close with a validation plan that requires per-basket timestamps the exported dataset does not contain.

Index Terms — association rules, Apriori, market-basket analysis, lift, confidence, recommender systems

1. Introduction

Problem statement.

- *Discovery of associations* — extract statements of the form "baskets containing A tend also to contain B" at support, confidence, and lift thresholds, retaining only rules with lift > 1 .
- *Confidence is easy to read and easy to game* — confidence ignores how frequent the consequent is, so rules into popular items pass any fixed floor almost automatically.
- *Threshold effects* — the number of frequent itemsets and rules is a steep, non-linear function of both sliders; lowering support $2\times$ multiplies rules $>10\times$.
- *Association is not causation* — a mined rule is observational co-occurrence; it does not license a causal claim about a promotion.
- *Scale* — 17,080 baskets over 3,653 items admit far more candidate itemsets than are frequent, so the miner must prune before it counts.

Motivation. The fourth assignment of an AI-assisted Recommender Systems course, moving from the personalised similarity signal of Week 3 to the co-occurrence signal: association rules are the machinery behind every "frequently bought together" widget, and they need no user model at all.

Concrete example. The user keeps the default sliders (support 1.0%, confidence 30%), presses **Run rules**, and gets 950 rules; clicking GREEN REGENCY TEACUP AND SAUCER $\rightarrow$ ROSES REGENCY TEACUP AND SAUCER opens the detail panel (support 3.17%, confidence 78.29%, lift 17.1440), and **Reverse direction** drops confidence to 69.36% while lift stays 17.1440.

Contributions.

- A working recommender in `index.html`, `style.css`, `script.js`, `transactions.js`, no runtime framework: basket construction, itemset counting, the three metrics with zero-denominator guards, a level-wise Apriori miner over posting lists, and rule generation in both directions — the seven `TODO(hw4)` stubs of the starter.
- Verified results: the built-in harness goes from 2 passed / 9 pending to **11 passed / 0 failed**; a hand-computed rule matches the page exactly; and all 1,200 exported rule rows were re-derived from raw counts with 0 discrepancies (Section 4, *Verification*).
- The mandatory useful-vs-misleading rule comparison, the optional direction analysis, a six-setting threshold sweep, and a validation plan that states exactly what data the starter does not contain.

2. Related Work

Prior work consulted: the UCI *Online Retail* dataset and its case study [1] [2], the original Apriori papers [3] [4], and background on association rule learning [5]. Full references are listed in the References section.

Alternatives considered. FP-growth would mine the 0.5%-support setting faster, but the brief specifies Apriori, and intersecting per-item posting lists instead of re-scanning baskets already keeps the 5,255-itemset run interactive in the browser. Ranking rules by confidence alone was rejected — Results (B) shows a rule that clears a 30% confidence floor while sitting at the mathematical minimum of its own lift. Personalised collaborative filtering (the Week 3 approach) was not applicable: a "frequently bought together" slot must work for an anonymous shopper with no rating history, which is exactly the non-personalised regime association rules occupy.

3. Method

Approach. The dataset is UCI *Online Retail* (dataset id 352) [1], a transactional log of a UK-based non-store online retailer covering 01/12/2010–09/12/2011 [2]. The starter export (produced by the instructor's cleaning pipeline, summarised in `assignment.md` §3) removes cancellations, non-positive quantities and prices, blank descriptions, guest checkouts, and non-product codes, yielding the frame of Table 1.

Table 1: dataset after the starter cleaning.

Property	Value
Baskets (invoices, ≥ 2 distinct items)	17,080
Distinct items (stock codes)	3,653
Kept rows	384,911 (from 541,909 raw)
Per-basket timestamp	absent (grouping key is InvoiceNo only)

The single most frequent item is 85123A — WHITE HANGING HEART T-LIGHT HOLDER — present in **1,959 of 17,080 baskets = 11.47%**; next are 22423 REGENCY CAKESTAND 3 TIER (9.84%) and 85099B JUMBO BAG RED RETROSPOT (9.27%). No item appears in more than ~11.5% of baskets — the fact that drives the misleading-rule analysis in Results (B). `dedupeBasket` turns each InvoiceNo into one transaction and collapses repeated stock codes (first-appearance order), so a basket is a *set*: an invoice listing 30 units of one SKU counts once, and support means "share of baskets". For itemset X, $\text{count}(X)$ is the number of baskets containing all of X, and $N = 17,080$; the metrics follow assignment §5:

```
support(A → B)    = count(A ∪ B) / N
confidence(A → B) = count(A ∪ B) / count(A)
lift(A → B)        = confidence(A → B) / [ count(B) / N ]
```

Support is symmetric, confidence is not, and lift cancels the direction: $\text{lift}(A \rightarrow B) = N \cdot \text{count}(A \cup B) / (\text{count}(A) \cdot \text{count}(B))$.

Pipeline. `dedupeBasket` builds the basket index; `countItemset` intersects sorted posting lists; `findFrequentItemsets` runs level-wise Apriori with the downward-closure property — a candidate k -itemset with any infrequent $(k-1)$ -subset is pruned before counting [3] [4]; `generateRules` splits every frequent itemset into antecedent/consequent in both directions (bitmask enumeration over all non-empty proper subsets), filters by confidence, and computes lift against the global N . The page then applies the two slider thresholds, renders a sortable results table (click a column header to sort; click again to flip direction), and opens a detail panel per rule with a **Reverse direction** button. Each metric returns `{ value, defined }` and reports `defined = false` on a zero denominator, so the page shows an inline note instead of NaN (assignment §4.1 item 8):

```
function computeLift(confidence, consequentCount, n) {
  if (!confidence || !confidence.defined) return { value: 0, defined: false };
  if (n === 0 || consequentCount === 0) return { value: 0, defined: false };
  return { value: confidence.value / (consequentCount / n), defined: true };
}
```

Tools & libraries. HTML5 / CSS3 / vanilla ES6, no runtime framework; UCI Online Retail [1] [2]; a local Python script driving the recomputation cross-check of the Verification section (deliberately not part of the submission — the deliverable contains no Python); Git / gh for deployment.

Key design decisions.

- *Sets, not multisets* — repeated units inside an invoice are collapsed, so support counts baskets, not quantity; the brief's support definition is "share of baskets".
- *Both directions, lift from global N* — `generateRules` emits $A \rightarrow B$ and $B \rightarrow A$ separately (their confidences differ), but each lift is computed against the same global item base rates, which is what makes the two lifts provably identical.
- *Thresholds: support $\geq 1\%$, confidence $\geq 30\%$, lift > 1 retained.* 1% support ≈ 171 baskets keeps every retained rule grounded in non-trivial volume while still admitting whole product families (herb markers, tea plates, lunch-bag lines). 30% confidence discards the long tail of one-off co-occurrences without demanding the near-deterministic 60%+ level, which keeps only 238 rules, mostly pairs from the same product pattern. The lift > 1 condition is applied on top of both; at this confidence floor it happens to be automatically satisfied (see *Comparison vs. baseline*), so the operational filter is really the two sliders. The empirical sweep around this operating point is in Results (C).
- *Evidence by export* — every quoted figure appears verbatim in the saved rule-table exports or is arithmetic shown in this report.

Configuration. PollyIva/Association-Rules-Recommender (public, main). Sliders: support 0.5–10% step 0.5 (default **1.0%**), confidence 10–100% step 5 (default **30%**); 950 rules at the defaults.

4. Experiments

Setup. The final rule table uses the thresholds stated and justified in *Key design decisions* (support $\geq 1\%$, confidence $\geq 30\%$, lift > 1 retained). For the threshold trade-off we additionally ran the miner at 0.5%/30%, 3%/30%, and 1%/60%. Full rule-table exports were saved for the three runnable settings (1%/30% $\rightarrow$ 950 rows, 3%/30% $\rightarrow$ 12 rows, 1%/60% $\rightarrow$ 238 rows); the 0.5%/30% run yields 10,723 rules, so it is cited by the page's status-line rule count instead of a copied table. Every figure quoted below appears verbatim in those exports or follows from the shown arithmetic. Page screenshots for both analysed rules (results table + selected-rule detail panel) are saved in the folders *useful rule/* and *weak rule/*.

Worked example. The hand-computation required by assignment §6.7, on real data: 23171 REGENCY TEA PLATE GREEN $\rightarrow$ 23172 REGENCY TEA PLATE PINK, taken from the index at minSupport = 1%:

```
count(A) = 269, count(B) = 224, count(A U B) = 202, N = 17,080

support    = 202 / 17080          = 0.011827  $\rightarrow$  1.18%
confidence = 202 / 269            = 0.750929  $\rightarrow$  75.09%
lift       = (202/269) / (224/17080)
           = 0.750929 / 0.013115 = 57.2584
```

The page shows exactly 1.18% / 75.09% / 57.2584. Reversing the direction: confidence = 202 / 224 = **90.18%** (changed), lift = 57.2584 (identical) — the direction-asymmetry of confidence and symmetry of lift, demonstrated on raw counts.

Results (A). One rule worth investigating, at the final thresholds:

Table 2: useful rule (spec §4.2 item 1).

Rule	count(AUB)	count(A)	count(B)	Support	Confidence	Lift
GREEN REGENCY TEACUP AND SAUCER $\rightarrow$ ROSES REGENCY TEACUP AND SAUCER	541	691	780	3.17%	78.29%	17.1440
(reverse) ROSES $\rightarrow$ GREEN	541	780	691	3.17%	69.36%	17.1440

Business action. Both items are pieces of the same "Regency" china pattern, and 541 baskets (~1 in 31 of all orders) already contain both. I would ship this as (i) a "Complete the set" cross-sell slot on both product-detail pages and (ii) a two-item bundle with a small discount. The antecedent count, 691 baskets (4.05% of all orders), is large enough that even a modest attach-rate uplift moves total units.

Why lift, not confidence. Confidence 78.29% sounds impressive in isolation, but it must be read against the consequent's base rate: the ROSES teacup appears in only 780/17,080 = 4.57% of baskets. Buyers of the GREEN cup are therefore ~17 $\times$ more likely to take a ROSES cup than a random basket (lift 17.1440). Had we ranked rules by confidence alone, rules whose consequents are merely frequent items would float up (Results (B)); lift corrects for exactly that. The symmetric figure confirms the pairing is a genuine co-purchase pattern rather than an artefact of one item's popularity: the reverse direction keeps the same lift while its confidence differs (78.29% vs 69.36%).

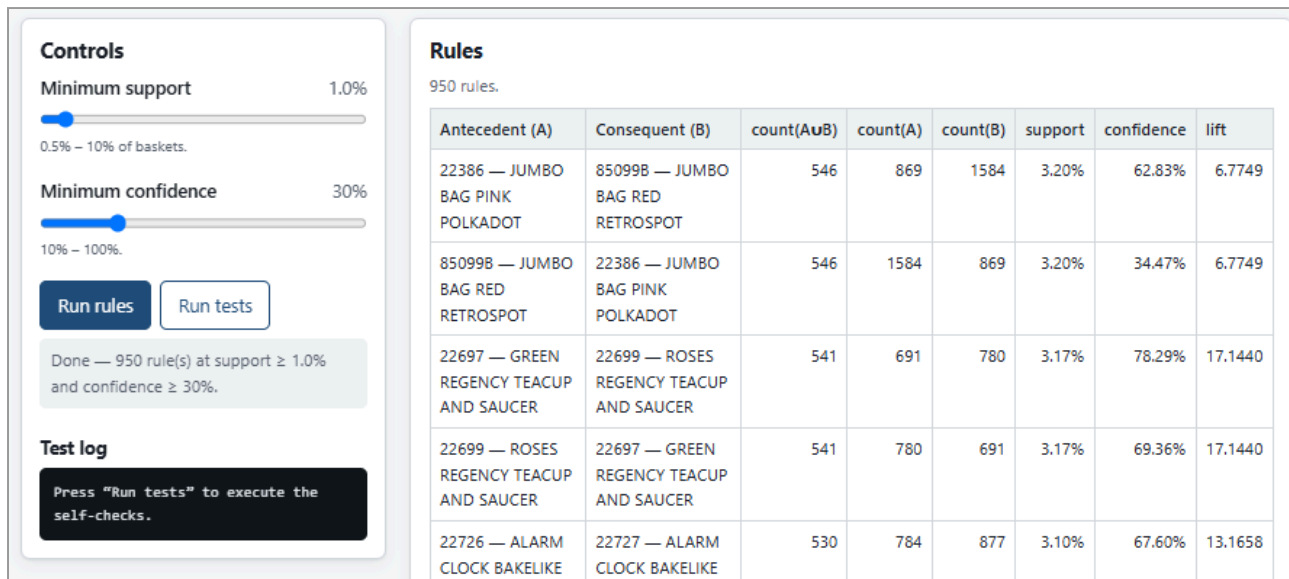


Fig. — Run rules at support 1% / confidence 30% — the teacup rule in the results table

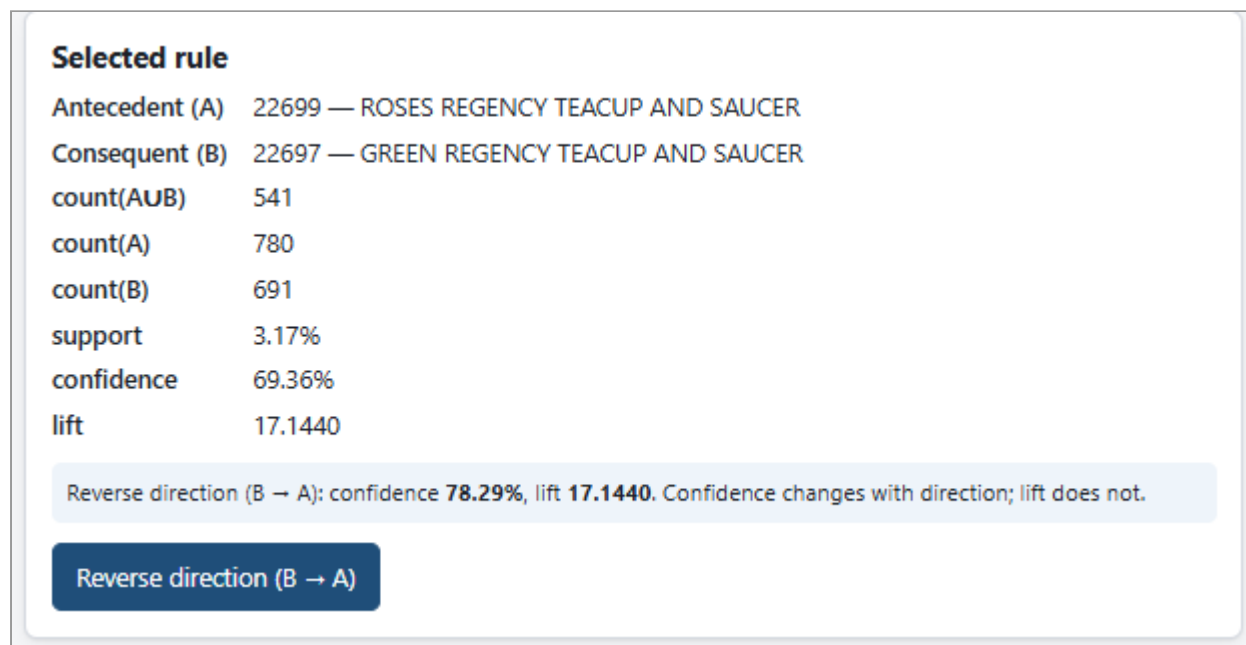


Fig. — Selected-rule detail panel, reverse direction selected: ROSES → GREEN, confidence 69.36%, lift 17.1440, with the note that the other direction gives 78.29% and the same lift

Results (B). One rule to reject (misleading, spec §4.2 item 2). The assignment's anchor fact is that 85123A — WHITE HANGING HEART T-LIGHT HOLDER — occurs in 11.47% of baskets. Fittingly, the five lowest-lift rules of the whole 950-rule table all have 85123A as the consequent. The second-lowest (2.6409; the absolute minimum, 2.6366, is 23321 SMALL WHITE HEART OF WICKER → 85123A) is:

Table 3: misleading rule.

Rule	count(AuB)	count(A)	count(B)	Support	Confidence	Lift
HOME BUILDING BLOCK WORD → WHITE HANGING HEART T-LIGHT HOLDER	209	690	1,959	1.22%	30.29%	2.6409

Why it misleads. A naive reading says "30% of block-word buyers also buy the white heart — a useful association." Two corrections apply:

1. *The confidence floor makes passing almost automatic.* For any rule whose consequent is 85123A, lift = confidence / 0.1147. At confidence $\geq 30\%$ the smallest achievable lift is $0.30 / 0.1147 = \mathbf{2.62}$, and lift > 1 is then guaranteed with room to spare. This rule scores 2.6409 — it sits within 1% of the mathematical floor of what the filter can admit. It did not clear a bar; it inherited the bar from the consequent's own popularity.
2. *The absolute volume is modest.* Support 1.22% = 209 baskets; the rule says nothing about *why* the two items co-occur and nothing beyond this observed period.

Controls

Minimum support0.5%

0.5% – 10% of baskets.

Minimum confidence30%

10% – 100%.

Run rules

Run tests

Done — 10723 rule(s) at support ≥ 0.5% and confidence ≥ 30%.

Test log

Press “Run tests” to execute the self-checks.

	CHARLOTTE BAG						
21754 — HOME BUILDING BLOCK WORD	85123A — WHITE HANGING HEART T-LIGHT HOLDER	209	690	1959	1.22%	30.29%	2.6409
22633 — HAND WARMER UNION JACK	22865 — HAND WARMER OWL DESIGN	209	441	571	1.22%	47.39%	14.1762
22865 — HAND WARMER OWL DESIGN	22633 — HAND WARMER UNION JACK	209	571	441	1.22%	36.60%	14.1762
23208 — LUNCH BAG VINTAGE LEAF DESIGN	23202 — JUMBO BAG VINTAGE LEAF	209	595	772	1.22%	35.13%	7.7714
21930 — JUMBO STORAGE BAG SKULLS	22386 — JUMBO BAG PINK POLKADOT	207	469	869	1.21%	44.14%	8.6749
22667 — RECIPE BOX RETROSPOT	22666 — RECIPE BOX PANTRY YELLOW DESIGN	206	439	826	1.21%	46.92%	9.7031

Fig. — Run rules at support 0.5% / confidence 30% — the weak rule in the results table

Selected rule	
Antecedent (A)	21754 — HOME BUILDING BLOCK WORD
Consequent (B)	85123A — WHITE HANGING HEART T-LIGHT HOLDER
count(AUB)	209
count(A)	690
count(B)	1959
support	1.22%
confidence	30.29%
lift	2.6409
Reverse direction (B → A): confidence 10.67% , lift 2.6409 . Confidence changes with direction; lift does not.	
Reverse direction (B → A)	

Fig. — Selected-rule detail panel: HOME BUILDING BLOCK WORD → WHITE HANGING HEART T-LIGHT HOLDER, counts 209/690/1959, support 1.22%, confidence 30.29%, lift 2.6409, reverse direction 10.67% with unchanged lift

Results (C). Threshold trade-off (spec §4.2 item 3). The measured effect of moving the two sliders around the operating point of *Key design decisions*:

Table 4: frequent itemsets and rules at six settings.

Support	Confidence	Frequent itemsets	Rules
0.5%	30%	5,255	10,723
1%	30%	1,219	950
3%	30%	115	12
1%	60%	1,219	238
1%	90%	1,219	18
3%	60%	115	5

Support acts before rule generation; confidence acts after. Lowering support from 1% to 0.5% grows the frequent-itemset set from 1,219 to 5,255 — 1-itemsets from 691 to 1,320, 2-itemsets from 413 to 2,206, and itemsets up to size 6 appear (sizes at 1%: 691/413/109/6; at 0.5%: 1320/2206/1161/474/88/6). Because of downward closure, each newly admitted 1-item unlocks its pairs, each frequent pair unlocks triples, and the candidate space compounds — hence the

10× growth in both itemsets and rules (10,723) for a 2× loosening of the threshold. Raising support to 3% reverses it: only 109 single items and 6 pairs survive, so only 12 rules can be generated at all. Confidence does not touch the itemset count (1,219 itemsets at both 30% and 60%) — it only filters the rules already generated: 950 → 238 → 18 as the confidence floor rises 30% → 60% → 90%. The two knobs act at different stages of the pipeline, which is why the table separates them.

Rule direction (optional, spec §4.2 counterpart). Every frequent pair generates two candidate rules. Three pairs at the final thresholds (all figures from the saved exports):

Table 5: confidence is directional, lift is not.

$A \rightarrow B$	$\text{conf}(A \rightarrow B)$	$\text{conf}(B \rightarrow A)$	lift (identical both ways)
GREEN REGENCY TEACUP → ROSES REGENCY TEACUP	78.29% (541/691)	69.36% (541/780)	17.1440
REGENCY TEA PLATE GREEN → REGENCY TEA PLATE PINK	75.09% (202/269)	90.18% (202/224)	57.2584
JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT	62.83% (546/869)	34.47% (546/1584)	6.7749

Both directions share the numerator $\text{count}(A \cup B)$; only the denominator changes, and whenever $\text{count}(A) \neq \text{count}(B)$ — almost always — the confidences differ. The third row makes the size of the effect concrete: the pink polkadot bag appears in 869 baskets and the red retrospot bag in 1,584, so "given pink, red follows" holds in 62.83% of pink baskets while "given red, pink follows" holds in only 34.47% of red baskets — the more frequent item is the weaker antecedent, because knowing it tells you less. The lifts stay identical because $\text{lift}(A \rightarrow B) = N \cdot \text{count}(A \cup B) / (\text{count}(A) \cdot \text{count}(B))$ is symmetric. *Which direction to deploy:* the antecedent is the *trigger* — the rule fires when a shopper views item A, and its confidence is the expected attach rate of recommending B on A's page — so direction follows the merchandising context, while lift only tells you whether the pair is worth deploying at all.

Comparison vs. baseline. The baseline is independence, i.e. lift = 1: a rule whose consequent appears at the same rate inside the antecedent as everywhere. At the final operating point every rule clears it — not by much, but by arithmetic: with confidence $\geq 30\%$ and no consequent above 11.47% base rate, lift $\geq 0.30 / 0.1147 \approx 2.62$ for *every* admitted rule, which is why all 950 rules have lift > 1 and the lift filter removes nothing (the misleading rule of Results (B) sits exactly at this floor). The filter only becomes operative at lower floors: at confidence $\geq 10\%$ the page admits REGENCY CAKESTAND 3 TIER → WHITE HANGING HEART with confidence 13.92% but lift **1.2137** — co-occurrence barely above chance (13.9% vs the 11.47% base rate), the clearest example of confidence looking "positive" while lift says "independent". An offline re-mine over all 21,026 rules generated at support $\geq 0.5\%$ / confidence $\geq 0.5\%$ found exactly 12 rules with lift < 1 (minimum 0.9267): without the lift condition, a fixed confidence floor admits rules that are *anti*-correlated once the base rate is taken into account.

Verification. Four checks back every number above:

1. *Built-in self-tests.* The page's own harness reports the baseline 2 passed / 0 failed / 9 pending on the untouched stubs, and **11 passed / 0 failed / 0 pending** after implementation — including the fixture's lift = 1, lift > 1, lift < 1, reversed-confidence, and zero-denominator cases.
2. *Hand-computation against raw counts* (assignment §6.7): the worked example above matches the page exactly, in both directions.
3. *Cross-check of every exported row.* All 1,200 rule rows saved from the three exported settings were re-derived from their raw counts in an independent recomputation: support, confidence, and lift matched the exported values in **all 1,200 rows (0 discrepancies)**, every row satisfied its section's thresholds, and the declared header counts matched the actual row counts.
4. *Reproducibility.* The rule-table exports and the four screenshots are saved alongside the submission; every figure in this report appears in them or is an arithmetic combination shown above. The 0.5%/30% setting is cited by its status-line count (10,723) rather than a full-table export, which was impractical to copy.

5. Discussion

Failure case. The rule we deliberately keep as the counter-example: HOME BUILDING BLOCK WORD → WHITE HANGING HEART T-LIGHT HOLDER, 209/690/1959, support 1.22%, confidence 30.29%, lift 2.6409 (Table 3). It passes both sliders and the lift > 1 requirement, it is present in two exports and in the screenshots, and it is still worthless as a recommendation.

Root cause. Two facts compose. The consequent sits in 11.47% of all baskets, so any rule into it with confidence $\geq 30\%$ has lift ≥ 2.62 — the filter's bar is *given* by the item's own popularity, not by the strength of the pairing. And confidence is computed against count(A) alone, so it has no knowledge of count(B): 30.29% vs the 11.47% base rate is a lift of 2.64, i.e. the association is at the edge of what the floor can admit. The deeper cause of the pairing itself may be nothing but a wholesaler restocking both lines — baskets are invoices, not customer journeys.

Fix + verification. *Detection:* the harness's lift < 1 fixture and the floor arithmetic above — before trusting any rule we check its lift against the consequent's base rate rather than reading confidence alone. *Change:* rank and filter on lift, keep the lift > 1 condition as a guard for lower confidence floors, and prefer mid-support rules with lift far above the floor — Table 2's teacup rule (lift 17.14 at support 3.17%) is the shape to look for. *Confirmation:* 11/11 self-tests, the hand-computed worked example, and the 1,200-row cross-check (0 discrepancies) establish that the metrics behind that judgement are computed correctly; the misleading rule itself is present in the exports at 2.6409 exactly.

Association is not causation (spec §4.2 item 5). $A \rightarrow B$ states that A and B co-occur more often than independence predicts. It does **not** state that showing or promoting A *causes* B to be sold: the rule is computed from observational co-occurrence, with no intervention and no randomisation. For the teacup rule a plausible confounder is *product-line membership*: both SKUs are pieces of the same Regency pattern, and a third factor — customers buying the pattern (or a wholesaler restocking the whole range) — causes both purchases. Merchandising is another: if the two cups were displayed adjacent in the catalogue all along, the observed pairing reflects a pre-existing placement, so "promote $A \rightarrow B$ sells" confuses the effect of a policy that was never varied. The lift of 17.14 quantifies the association; it puts no bound on the incremental sales an intervention on A would produce.

Feature to ship and its most important limitation (spec §4.2 item 4). *Feature*: a "Frequently bought together" row on the product-detail pages of the Regency tea range — the GREEN/ROSES teacup pairing of Table 2 — with a one-click add-both basket button and shelf co-location of the two SKUs in the physical warehouse pick list. *Most important limitation*: the dataset mixes wholesale and retail demand (the UCI source notes that "many customers of the company are wholesalers" [1]), and baskets are invoices — a single wholesaler restocking an entire pattern range produces exactly this rule without any end-customer preference behind it. Compounding this, the export carries no per-basket timestamp, so the rule is a frequency statement about one observed period (Dec 2010–Dec 2011) only: it cannot separate a durable pairing from a seasonal gifting spike, and it will silently decay if the product range changes.

What worked.

- *Sets and posting lists* — deduping baskets kept support semantics honest, and intersecting sorted id lists made even the 5,255-itemset run interactive with no libraries.
- *Lift-first reading* — every rule we quote is presented with its consequent's base rate, so the difference between "30% confident" and "17× enriched" is visible on the page itself.
- *Evidence by export* — the three saved rule tables plus four screenshots let every claim in this report be checked against a file rather than trusted.

What surprised me.

- that the lift > 1 condition I was asked to apply turns out to be mathematically redundant at the operating point I chose — the 30% confidence floor already forces lift ≥ 2.62 on this dataset.
- that a 2× change in one threshold multiplies the rule count by more than 10× (950 $\rightarrow$ 10,723), and that the two sliders act at different pipeline stages, so they do not even have the same units of effect.

Next improvement. The validation plan (spec §4.2 item 6). The starter export groups baskets by InvoiceNo and **drops the date**; there is no per-basket timestamp, and the assignment forbids simulating one, so none of the following can be executed on the starter data — it is a specification for what to measure once timestamped or experimental data is available:

1. *Stability check on a later period* (when later-period baskets with timestamps exist): mine rules on period 1 (months 1–7), then *evaluate* — not re-mine — them on period 2 (months 8–9) by recomputing support, confidence, and lift from period-2 counts. Decision rule: keep a rule only if period-2 lift > 1 with a bootstrap 95% confidence interval excluding 1, and period-2 support $\geq 0.5\times$ its period-1 value; otherwise reject or re-tune.
2. *A/B test before deploying the bundle*: randomly assign product-page sessions to treatment (cross-sell slot showing the GREEN + ROSES pairing) or control (no slot). Primary metric: incremental attach rate, $P(\text{basket contains ROSES} \mid \text{session saw GREEN page, treatment})$ minus the same

quantity in control; secondary: basket size. Ship only if the one-sided test rejects H_0 at $\alpha = 0.05$ **and** the bundled pair's lift stays > 1 in the treatment arm.

3. *Smallest feasible experiment if no historical later period exists*: collect prospectively — log baskets with timestamps for a fixed window, mine on the first two-thirds, hold out the final third for the stability test of point 1, then run the A/B test of point 2 on the validated rules.

6. AI Usage Disclosure

AI tools used. OpenCode (AI coding CLI, model big-pickle) — categories: code generation, debugging, measurement assembly (the independent 1,200-row recomputation cross-check), documentation.

What I personally verified.

- Ran the page in a browser: executed the self-tests (11 passed / 0 failed), ran the miner at all cited settings, clicked through both analysed rules, used the Reverse direction button, and checked the sortable table headers.
- Double-checked every number in this report against the saved rule-table exports and screenshots, and re-did the worked example's arithmetic by hand.
- Chose which rules to present as useful vs. misleading, and confirmed the five lowest-lift claims by sorting the full 950-rule export.

References

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